

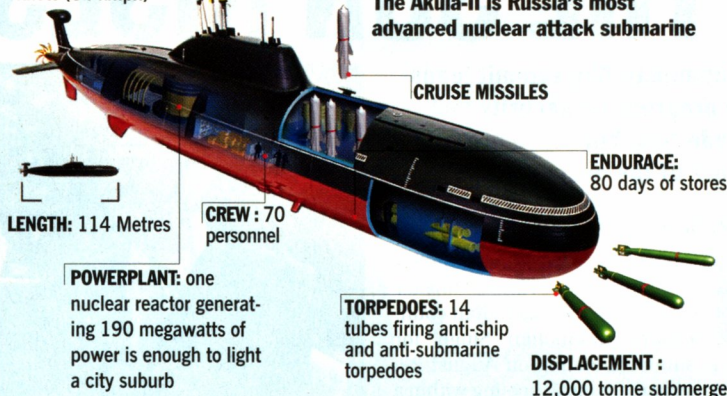
denied reports of the lease, but in 2005, the Russian daily *Kommersant* noted that the unaccounted for spike in the country's arms export earnings indicated that the lease had been paid up.

The *Chakra* will soon be joined by the indigenous ATV, under construction at a secret dry dock in Visakhapatnam. Construction of the 5,000-tonne ATV, a modified version of the Russian Charlie-II class is now nearly complete, and will be launched for sea trials next year. It will be inducted into the Indian Navy in 2009. Armed with indigenously developed ballistic missiles (future variants with the three-stage 5000-km range Agni 3), the ATV will mark India's entry into the SSBN club and will mean the fruition of a long-delayed strategic programme.

The delays seem to have moved to the Russian side. Originally slated for induction on August 15 this year, the delivery of the *Chakra* has been delayed by 10 months for the same reasons that delayed the Gorshkov refit in Russia. Earlier this year, Russia escalated the cost of the N sub lease by \$135 million (Rs 540 crore), which was rejected by the Indian Defence Ministry delegation. Ministry officials confirmed the advanced stage of both the lease and the projects and said that the Government was debating on when to bring both the programmes out of the closet.

The lease of the Akula-II submarine—originally slated for the cash-

MAX SPEED : 35
knots (64 kmph)



UNDERSEA PREDATOR

The Akula-II is Russia's most advanced nuclear attack submarine

strapped Russian Navy and on which construction had ceased at the Amur shipyard in the 1990s—will make India the world's sixth power to operate a nuclear submarine. It has only one precedent—the three-year transfer of a Charlie-I class nuclear attack submarine (also named *Chakra*) from the Soviet Union in January 1988, which took advantage of a loophole in international treaties. The treaties prohibit the sale of nuclear submarines but do not object to a lease, provided the submarines are not equipped with nuclear weapons or missiles with a range of over 300 km. The *Chakra* will be stripped of its inventory of strategic cruise missiles with a range of 3,000 km, as these violate the Missile Technology Control Regime, but India will not be prevented from equipping the submarine with its own missiles.

The present 10-year lease—which may be extended later—differs from that of the Charlie-I class submarine in some im-

portant aspects. While the latter's reactor controls and missile launch area were manned by Soviet naval personnel, the new *Chakra* will be manned entirely by an Indian crew, which is to leave for Vladivostok in December. Nearly 300 Indian naval personnel, or three sets of crews, have already been trained to man the submarine at a specially constructed facility in Sosnovy Bor, a small town near St Petersburg in Russia. All personnel returned after completion of training this year.

Future ATV crews will also be trained on the *Chakra*, which offers a valuable training platform. "A leased submarine gives you a tremendous headstart in training crews," says Menon. "It takes several years to produce a crew of nuclear submarine experts like hydroplane operators and watch keeping officers." The new *Chakra* will make up for the expertise that was lost when the Charlie-I submarine was returned to the former Soviet Union but also add a strategic platform into India's inventory. ■

